PHD125 Pharmaceutical Organic Chemistry · Saturated and Unsaturated Hydrocarbons

After class: check your understanding

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 30 minutes. Questions follow the two subchapters of the lecture and end with a pharmacy scenario.

Answer without notes first, then use the slides to check. The answer key is at the back.

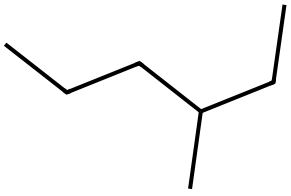
2.1 · Alkanes: structure and names

Q1 CH2-LO1

What is the molecular formula of the member of the alkane series just above C_8H_{18}? Type it on one line, e.g. C4H10. (Ball 12.E)

Q2 CH2-LO1

Name this compound.



A. 3-methylhexane

B. 4-methylhexane

C. 2-ethylpentane

D. heptane

Q3 CH2-LO1

Name $CH_3CH(CH_3)CH_2CH_2CH(CH_3)CH_2CH_3$.

A. 2,5-dimethylheptane
B. 3,6-dimethylheptane
C. 2,5-methylheptane
D. 2-methyl-5-methylheptane

Q4 CH2-LO1

Which name is **incorrect**?

A. 2-ethylbutane
B. 3-methylpentane
C. 2,2-dimethylpropane
D. 2-methylbutane

Q5 CH2-LO1

How many carbon atoms are in 3-ethyl-2-methylheptane? (Ball 12.E)

_____

Q6 CH2-LO1

$CH_3CH_2CH_2CH_3$ and a bent drawing of the same four-carbon chain are isomers.

True / False

Q7 CH2-LO1

Match each cycloalkane to its formula.

Options: C_3H_6 · C_5H_{10} · C_7H_{14} · C_6H_{14}

cyclopropane _____

cyclopentane _____

methylcyclohexane _____

2.1 · Alkanes: properties and reactions

Q8 CH2-LO2

25 mL of hexane is added to 100 mL of water. What happens? (Ball 12.E)

A. Hexane dissolves

B. Hexane does not dissolve and floats on top

C. Hexane does not dissolve and sinks

D. The two react

Q9 CH2-LO2

Which member of each pair has the higher boiling point? (Ball 12.E)

Options: first · second

pentane or butane _____

heptane or nonane _____

pentane or 2,2-dimethylpropane _____

Q10 CH2-LO3

Balance: C_4H_{10} + __ O_2 → 4 CO_2 + 5 H_2O. Give the coefficient of O_2 as a number or fraction.

_____

Q11 CH2-LO3

Methane and chlorine are mixed and kept in the dark at room temperature. What happens?

A. CH_3Cl forms quickly

B. Nothing: UV light or heat is needed

C. An explosion

D. CCl_4 forms

Q12 CH2-LO3

Match each step to its stage in the chlorination of methane.

Options: initiation · propagation · termination

$Cl_2 \rightarrow 2\ Cl\bullet$ _____

$Cl\bullet + CH_4 \rightarrow \bullet CH_3 + HCl$ _____

$\bullet CH_3 + Cl\bullet \rightarrow CH_3Cl$ _____

Q13 CH2-LO3

Which halogen reacts least readily with alkanes? (Ball 12.E)

A. F_2

B. Cl_2

C. Br_2

D. I_2

2.2 · Alkenes, alkynes and aromatics

Q14 CH2-LO4

Name $CH_3CH_2CH_2CH_2CH_2CH{=}CHCH_3$. (FGOB 13.2)

A. 2-octene

B. 6-octene

C. 2-heptene

D. octane

Q15 CH2-LO4

Which can exist as cis-trans isomers?

A. $CH_2{=}CHCH_2CH_2CH_3$

B. $CH_3CH{=}CHCH_2CH_3$

C. $(CH_3)_2C{=}CHCH_3$

D. $CH_2{=}CH_2$

Q16 CH2-LO4

Which carbon in $CH_3CH(Cl)CH_2CH_3$ is chiral?

A. C1
B. C2
C. C3
D. none

Q17 CH2-LO5

Match each reagent with the product it gives with propene, $CH_2{=}CHCH_3$.

Options: propane · 1,2-dibromopropane · 2-bromopropane · 2-propanol · 1-propanol

H_2, Ni _____

Br_2 _____

HBr _____

H_2O, H_2SO_4 _____

Q18 CH2-LO5

A few drops of red-brown bromine are added to two liquids. Liquid X stays red-brown; liquid Y turns colourless at once. Which is most likely?

A. X is cyclohexene, Y is hexane
B. X is hexane, Y is cyclohexene
C. Both are alkenes
D. Both are alkanes

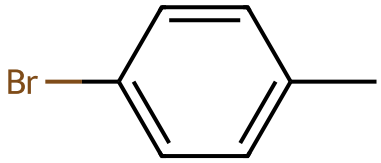
Q19 CH2-LO5

Which monomer makes PVC, used for IV fluid bags?

A. $CH_2{=}CH_2$
B. $CH_2{=}CHCH_3$
C. $CH_2{=}CHCl$
D. $CF_2{=}CF_2$

Q20 CH2-LO4

Name this compound.



A. p-bromotoluene (4-bromotoluene)
B. m-bromotoluene
C. o-bromotoluene
D. 1-bromo-3-methylbenzene

Q21 CH2-LO5

Benzene reacts with Br_2 only when $FeBr_3$ is present. The products are:

A. 1,2-dibromocyclohexane
B. bromobenzene + HBr
C. bromocyclohexane
D. no reaction ever

Pharmacy scenario · The ointment bench

Scenario

In the compounding room, you prepare an ointment using white soft paraffin as the base. A colleague suggests replacing it with olive oil "because it is natural". Olive oil is mostly triglycerides of oleic acid, which contains a **cis** C=C double bond. The label on the paraffin says *store below 25 °C, keep away from flames*.

Q22 CH2-LO2

Why is white soft paraffin a good ointment base?

A. It dissolves in water and washes off easily

B. It is a nonpolar, unreactive alkane mixture that forms a water-repelling layer on the skin

C. It contains a benzene ring

D. It is ionic

Q23 CH2-LO5

Olive oil slowly goes rancid in air; paraffin does not. Which structural feature explains this?

A. Olive oil's C=C double bonds can react (for example with oxygen)

B. Paraffin has more double bonds

C. Olive oil is an alkane

D. Paraffin is aromatic

Q24 CH2-LO3

Why does the paraffin label say "keep away from flames"? Write the type of reaction and its products.

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Adapted from LibreTexts *Introduction to Organic Chemistry – Alkanes* 12.3, 12.6–12.10 and exercises 12.E (Sharpe Elles, from Ball, Hill & Scott), *The Basics of GOB Chemistry* (Ball, Hill & Scott) chapter 13 and *Fundamentals of General Organic and Biological Chemistry* 13.1–13.9, CC BY-NC-SA 3.0. Reaction mechanisms, Markovnikov items and pharmacy scenarios are original material by Dr. NAM. Shared under CC BY-NC-SA for non-commercial teaching.

PHD125 Pharmaceutical Organic Chemistry · Saturated and Unsaturated Hydrocarbons

After class: check your understanding

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About 30 minutes. Questions follow the two subchapters of the lecture and end with a pharmacy scenario.

Answer key

2.1 · Alkanes: structure and names

Q1 CH2-LO1

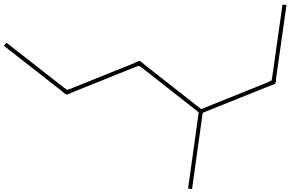
What is the molecular formula of the member of the alkane series just above C_8H_{18}? Type it on one line, e.g. C4H10. (Ball 12.E)

C_9H_{20}

Add one CH_2: C_9H_{20} (nonane).

Q2 CH2-LO1

Name this compound.



A. 3-methylhexane

B. 4-methylhexane

C. 2-ethylpentane

D. heptane

Longest chain 6 C; number from the end nearer the branch: C3. (Ball 12.E 6a)

Q3 CH2-LO1

Name $CH_3CH(CH_3)CH_2CH_2CH(CH_3)CH_2CH_3$.

A. 2,5-dimethylheptane

B. 3,6-dimethylheptane

C. 2,5-methylheptane

D. 2-methyl-5-methylheptane

7-carbon chain; numbering from the left gives 2 and 5 (from the right, 3 and 6). Use di- for two methyls. (Ball 12.E 6b)

Q4 CH2-LO1

Which name is **incorrect**?

A. 2-ethylbutane

B. 3-methylpentane

C. 2,2-dimethylpropane

D. 2-methylbutane

In "2-ethylbutane" the ethyl extends the chain to five carbons: the correct name is 3-methylpentane.

Q5 CH2-LO1

How many carbon atoms are in 3-ethyl-2-methylheptane? (Ball 12.E)

10 C

7 (heptane) + 2 (ethyl) + 1 (methyl) = 10.

Q6 CH2-LO1

$CH_3CH_2CH_2CH_3$ and a bent drawing of the same four-carbon chain are isomers.

True / False → **False**

Bending is rotation about C–C bonds: same compound (Ball 12.E 5a).

Q7 CH2-LO1

Match each cycloalkane to its formula.

cyclopropane → C_3H_6; cyclopentane → C_5H_{10}; methylcyclohexane → C_7H_{14}

Cycloalkanes are C_nH_{2n}. C_6H_{14} is hexane, an open chain.

2.1 · Alkanes: properties and reactions

Q8 CH2-LO2

25 mL of hexane is added to 100 mL of water. What happens? (Ball 12.E)

A. Hexane dissolves

B. Hexane does not dissolve and floats on top

C. Hexane does not dissolve and sinks

D. The two react

Hexane is nonpolar (insoluble) and less dense than water (0.66 g/mL).

Q9 CH2-LO2

Which member of each pair has the higher boiling point? (Ball 12.E)

pentane or butane → first; heptane or nonane → second; pentane or 2,2-dimethylpropane → first

Longer chains boil higher; branching lowers the boiling point.

Q10 CH2-LO3

Balance: C_4H_{10} + __ O_2 → 4 CO_2 + 5 H_2O. Give the coefficient of O_2 as a number or fraction.

6.5

O atoms on the right: 8 + 5 = 13, so 13/2 O_2. Doubled: 2 C_4H_{10} + 13 O_2 → 8 CO_2 + 10 H_2O.

Q11 CH2-LO3

Methane and chlorine are mixed and kept in the dark at room temperature. What happens?

A. CH_3Cl forms quickly

B. Nothing: UV light or heat is needed

C. An explosion

D. CCl_4 forms

Halogenation needs UV light or high temperature to split Cl_2 into radicals.

Q12 CH2-LO3

Match each step to its stage in the chlorination of methane.

Cl_2 → 2 Cl• → initiation; Cl• + CH_4 → •CH_3 + HCl → propagation; •CH_3 + Cl• → CH_3Cl → termination

Initiation makes radicals; propagation keeps the chain going; termination removes radicals.

Q13 CH2-LO3

Which halogen reacts least readily with alkanes? (Ball 12.E)

A. F_2

B. Cl_2

C. Br_2

D. I_2

F_2 reacts explosively; I_2 is too unreactive.

2.2 · Alkenes, alkynes and aromatics

Q14 CH2-LO4

Name $CH_3CH_2CH_2CH_2CH_2CH{=}CHCH_3$. (FGOB 13.2)

A. 2-octene

B. 6-octene

C. 2-heptene

D. octane

8-carbon chain; number from the right so the C=C starts at C2.

Q15 CH2-LO4

Which can exist as cis-trans isomers?

A. $CH_2{=}CHCH_2CH_2CH_3$

B. $CH_3CH{=}CHCH_2CH_3$

C. $(CH_3)_2C{=}CHCH_3$

D. $CH_2{=}CH_2$

Each C=C carbon needs two different groups. 2-Pentene has H and CH_3 on one, H and C_2H_5 on the other. (FGOB 13.3)

Q16 CH2-LO4

Which carbon in $CH_3CH(Cl)CH_2CH_3$ is chiral?

A. C1

B. C2

C. C3

D. none

C2 carries H, Cl, CH_3 and CH_2CH_3: four different groups.

Q17 CH2-LO5

Match each reagent with the product it gives with propene, $CH_2{=}CHCH_3$.

H_2, Ni → propane; Br_2 → 1,2-dibromopropane; HBr → 2-bromopropane; H_2O, H_2SO_4 → 2-propanol

HBr and H_2O follow Markovnikov's rule; 1-propanol is the minor product.

Q18 CH2-LO5

A few drops of red-brown bromine are added to two liquids. Liquid X stays red-brown; liquid Y turns colourless at once. Which is most likely?

A. X is cyclohexene, Y is hexane

B. X is hexane, Y is cyclohexene

C. Both are alkenes

D. Both are alkanes

Alkenes add Br_2 and decolourise it; alkanes do not react in the dark.

Q19 CH2-LO5

Which monomer makes PVC, used for IV fluid bags?

A. $CH_2=CH_2$

B. $CH_2=CHCH_3$

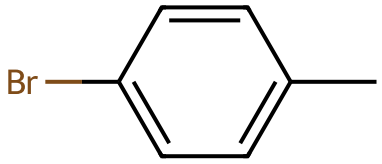
C. $CH_2=CHCl$

D. $CF_2=CF_2$

Vinyl chloride (chloroethene) → polyvinyl chloride. (FGOB 13.7)

Q20 CH2-LO4

Name this compound.



A. p-bromotoluene (4-bromotoluene)

B. m-bromotoluene

C. o-bromotoluene

D. 1-bromo-3-methylbenzene

Br is opposite (1,4) to the CH_3: para. (FGOB 13.9)

Q21 CH2-LO5

Benzene reacts with Br_2 only when $FeBr_3$ is present. The products are:

A. 1,2-dibromocyclohexane

B. bromobenzene + HBr

C. bromocyclohexane

D. no reaction ever

Benzene undergoes substitution, keeping its aromatic ring; HBr is the by-product.

Pharmacy scenario · The ointment bench

Q22 CH2-LO2

Why is white soft paraffin a good ointment base?

A. It dissolves in water and washes off easily

B. It is a nonpolar, unreactive alkane mixture that forms a water-repelling layer on the skin

C. It contains a benzene ring

D. It is ionic

Alkanes are inert and nonpolar, so they protect the skin and do not react with most drugs.

Q23 CH2-LO5

Olive oil slowly goes rancid in air; paraffin does not. Which structural feature explains this?

A. Olive oil's C=C double bonds can react (for example with oxygen)

B. Paraffin has more double bonds

C. Olive oil is an alkane

D. Paraffin is aromatic

Unsaturated C=C bonds are reactive sites. Saturated alkanes have none.

Q24 CH2-LO3

Why does the paraffin label say "keep away from flames"? Write the type of reaction and its products.

Alkanes are flammable: they undergo combustion. With enough oxygen: CO_2 + H_2O + heat; with limited oxygen: CO and soot too.

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